

Gabrielle Johnson's Safety Consideration

HCI — nurse decision-support handheld

1. Alarm fatigue from urgent ESI categorisation

Mechanism. The cost-sensitive weighting that cut ESI 2 under-triage from 34.6 % to 16.1 % did so by lowering precision. ESI 2 precision fell from ~0.74 to 0.602, hence roughly 1,988 of the ~4,996 patients flagged ESI 2 were over triaged. Triage staff may therefore be inclined to dismiss urgent alerts.

Mitigation. Disagreement is shown passively on the review screen and never triggers an interruption. Only deterioration while the patient is waiting may generate an interrupting alert. If precautionary flags exceed the agreed threshold, the class weights should be re-tuned.

2. Display Legibility

Mechanism. The handheld device screen is approximately 360×740 pixels and is typically used while standing, wearing gloves, and under bright overhead lighting, all within a three-to five-minute assessment. Only two or three SHAP features are visible above the fold. If the "Accept" button is reachable before contradictory evidence appears, the interface makes agreement physically easier than disagreement. This asymmetry can cause harm as key model caveats may remain below the fold, and a nurse who reviews only the supporting features may never see the warnings

Mitigation. Layout ordering is pivotal for safety. Contradicting features, low confidence levels and the blind-spot card must render above the accept button, so the nurse must scroll past the counter arguments to agree and should be legible at first glance.

3. State loss due to interruptions

Mechanism. Triage nurses are interrupted constantly, if a nurse is pulled away mid-assessment and the interface discards entered vitals she may run the risk of re-entering and wasting crucial time, or the patient silently reverts to "unreviewed" with nobody "owning" them.

Mitigation. Assessment state is persisted continuously, so a paused review resumes at the exact field it stopped. Any patient who has checked in but not been reviewed carries a countdown and on expiry the case escalates to the charge nurse.

HRI — check-in kiosk and patient-worn watch

4. Silent sensor failure-creating a false impression of vigilance

Mechanism. A strap loosened by a restless patient, cold hands, poor perfusion or tremor all cause optical HR and SpO₂ hence producing unreliable readings. Therefore, staff who believe every waiting patient is monitored will scan the room less carefully, so an unmonitored deteriorating patient is now less visible than before the system existed.

Mitigation. Graceful degradation with an explicit failure signal: watches self-test and report their own status into the nurse's queue view, and a watch that stops reporting raises its own alert rather than falling quietly silent. The nurse's visual sweep of the waiting room is explicitly retained as the watch is an assistive tool.

5. Physical harm from watch

Mechanism. If the device is put on the wrong wrist, it could press on a dialysis fistula, damage a dressing, or hurt fragile skin. A patient who is confused, intoxicated, or distressed may not be able to take it off or say that it hurts. Moreover, if the same device is used on different patients, the strap could also transmit diseases to already vulnerable patients .

Mitigation. A contraindication check runs at issue, the strap opens by hand without a tool or a release code, and staff apply and remove it when necessary. The strap must be made from material that is hypoallergenic and can be sterilized, and must have a size range so it can be adjusted for patient comfort.

6. Public alert disclosing clinical status

Mechanism. A device that signals deterioration visibly tells strangers something about this person's condition. Moreover, fear raises heart and respiratory rate, worsening the very signal that triggered the alert, and a hyperventilating frightened patient can also generate a second false alarm

Mitigation. The public signal is a steady amber "needs staff attention" paired with calm non-diagnostic wording ("a nurse is coming to check on you"). All clinical details, the trend that fired the alert and the patient's location are sent privately to the nurse's handheld with a named owner who must acknowledge it and a timer.